

DRAVAGEN, Sweden

WMO: 023170

Lat: 62.094N

Lon: 13.609E

Elev: 566

StdP: 94.71

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.1	(c) -24.0	(d) -29.8	(e) 0.3	(f) -27.0	(g) -26.5	(h) 0.4	(i) -23.9	(j) 9.9	(k) 0.4	(l) 8.5	(m) -0.9	(n) 1.0	(o) 250	(p) 0.691

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	23.8	15.1	22.0	14.6	20.1	13.7	16.7	21.5	15.7	19.9	14.6	18.4	2.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	11.3	17.9	13.9	10.6	17.2	12.9	9.9	16.2	49.0	21.8	45.9	19.9	42.6	18.4	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.1	(b) 6.8	(c) 5.9	(e) -30.6	(f) 26.0	(g) 2.2	(h) 1.9	(i) -32.2	(j) 27.3	(k) -33.5	(l) 28.4	(m) -34.7	(n) 29.4	(o) -36.3	(p) 30.8		
			DB	WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-8.9	-7.8	-4.9	0.7	6.3	10.4	13.1	11.4	7.1	1.7	-3.5	-7.3
	DBStd	8.93	6.54	5.95	5.40	3.74	3.72	3.08	2.87	2.82	3.13	3.83	4.89	6.64
	HDD10.0	3302	586	499	461	280	126	32	6	17	94	259	404	537
	HDD18.3	6120	844	732	719	530	373	239	164	216	336	517	654	795
	CDD10.0	223	0	0	0	0	12	43	101	60	8	1	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	69	0	0	0	0	3	13	46	7	0	0	0	0
	CDH26.7	3	0	0	0	0	0	1	2	0	0	0	0	0
(14) Wind	WSAvg	2.4	2.4	2.4	2.8	2.7	2.7	2.7	2.2	2.1	2.3	2.4	2.2	2.3
(15) Precipitation	PrecAvg	605	31	25	25	32	49	74	90	84	61	54	43	35
	PrecMax	779	56	51	48	71	96	149	157	163	112	123	118	54
	PrecMin	472	6	8	6	7	9	18	47	44	27	12	19	11
	PrecStd	86	15	12	12	17	23	30	29	32	27	31	23	13
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	5.0	8.5	15.3	23.5	25.1	26.9	24.3	19.1	13.9	8.9	5.2
		MCWB	2.5	2.2	4.0	8.3	12.7	14.0	17.2	15.9	12.6	10.4	7.6	3.7
	2%	DB	3.0	3.7	6.2	12.4	20.5	22.6	24.6	22.1	17.0	11.2	6.4	4.0
		MCWB	1.7	1.7	3.0	6.4	12.1	13.5	16.4	15.2	12.1	8.6	5.3	2.6
	5%	DB	1.7	2.3	4.4	10.0	17.6	20.3	22.5	20.1	15.0	9.6	4.4	2.9
		MCWB	0.6	0.8	1.9	5.3	10.6	12.9	15.4	14.2	11.4	7.4	3.6	1.7
	10%	DB	0.2	0.7	2.9	7.6	14.4	18.0	20.4	17.9	13.2	8.1	2.9	1.5
		MCWB	-0.8	-0.5	0.9	3.8	9.1	11.6	14.5	13.6	10.3	6.7	2.1	0.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	3.0	4.5	9.2	13.8	16.2	18.2	17.4	14.2	11.2	7.6	4.0
		MCDB	3.9	4.4	7.7	13.8	20.1	20.9	23.9	21.3	17.0	12.9	8.7	4.8
	2%	WB	1.9	1.9	3.3	7.0	12.6	14.6	17.2	16.2	12.9	9.1	5.3	2.7
		MCDB	2.9	3.3	5.7	11.7	18.7	19.6	22.8	19.8	15.9	10.8	6.3	3.7
	5%	WB	0.8	0.8	2.1	5.5	11.2	13.5	16.2	15.2	11.8	7.8	3.7	1.7
		MCDB	1.8	2.3	4.1	9.2	16.3	18.9	21.2	18.5	14.4	9.0	4.3	2.7
	10%	WB	-0.8	-0.4	1.0	4.2	9.6	12.4	15.2	14.1	10.7	6.7	2.2	0.5
		MCDB	0.1	0.7	2.6	7.1	14.1	17.2	19.4	17.2	12.9	8.2	2.8	1.4
(35) Mean Daily Temperature Range	MDBR		8.8	9.6	10.5	10.0	10.7	11.1	11.1	10.7	9.5	7.8	6.7	8.3
	5% DB	MCDBR	8.0	8.5	10.6	14.3	17.2	16.4	15.8	14.8	13.4	10.7	7.5	7.5
		MCWBR	7.2	7.1	7.8	8.6	9.2	8.3	8.1	8.3	8.8	8.0	6.7	6.8
	5% WB	MCDBR	7.5	7.7	9.4	13.1	15.1	13.8	14.2	12.1	11.7	9.1	7.1	7.5
		MCWBR	6.9	6.6	7.1	8.2	8.5	7.6	7.8	7.4	8.1	7.3	6.4	6.8
(40) Clear-Sky Solar Irradiance	taub		0.204	0.257	0.285	0.318	0.328	0.324	0.339	0.329	0.299	0.284	0.219	N/A
	taud		2.114	2.281	2.308	2.374	2.477	2.516	2.483	2.500	2.567	2.499	2.192	N/A
	Ebn at Noon		520	729	835	875	895	903	877	858	827	702	490	N/A
	Edh at Noon		32	59	83	96	95	93	94	85	66	49	29	N/A
(44) All-Sky Solar Radiation	RadAvg		0.25	0.88	2.40	4.12	4.73	5.14	4.69	3.63	2.33	1.03	0.32	0.13
	RadStd		0.03	0.15	0.22	0.37	0.40	0.44	0.58	0.33	0.27	0.16	0.07	0.01

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page